

Milestone 4: Final Results and Project Report

Predictive Shipment Delay Classification & Dynamic Safety-Stock Adjustment System

Using the USAID SCMS Delivery History Dataset

Repository: github.com/MA-Gondal/CIND-820-Capstone-Project



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1. Introduction, Research Questions and Contributions

This report is the final documentation of the CIND820 capstone project designed in Milestone 1, architected in Milestone 2, and implemented as a proof of concept in Milestone 3. The project builds a machine learning system that classifies incoming shipments in a global health supply chain into three delay-risk categories: on-time, at-risk (1–14 days late), and delayed (more than 14 days late), using only attributes available at order placement, and then operationalizes the predictions into a dynamic, shipment-specific safety-stock formula. The primary beneficiary is the logistics manager or procurement planner in a global health organization, who must decide, before dispatch, which shipments justify expedited freight, alternative sourcing, or a larger inventory buffer. All analysis uses the USAID SCMS Delivery History Dataset (10,324 shipment line items, 33 columns, period of 2006-2015, 43 destination countries).

This report moves beyond the Milestone 3 proof of concept in five ways, each tied to a weakness recorded in that report: a hyperparameter search of the champion model (motivated by the exhausted early-stopping budget), repeated stratified cross-validation for the champion–baseline comparison (motivated by the five-pair minimum of the earlier statistical test), a deterministic final-fit protocol (motivated by small run-to-run metric variation observed with multithreaded training), SHAP attribution to complete the feature-importance analysis, and a cost-sensitive decision layer plus the safety-stock simulation that closes the impact chain to the beneficiary.

1.1 Research Questions

RQ1. Which pre-shipment attributes: freight mode, destination country, vendor, commodity type, line-item value, and shipment weight, are the strongest predictors of whether a shipment will be delivered on time, at risk, or delayed relative to its scheduled delivery date?

RQ2. Does a gradient-boosted multi-class classifier (XGBoost) achieve meaningfully superior macro-averaged F1 and macro-averaged AUPRC compared to a logistic regression baseline on the three-class task, and is the difference statistically significant under paired non-parametric testing?

RQ3. Can the trained model's predicted delay probabilities be operationalized into a dynamic, shipment-specific safety-stock adjustment formula, $SS = z \times \sigma(\text{predicted lead-time deviation}) \times \text{average daily demand}$, that reduces stockout exposure relative to a fixed-buffer policy without proportionally increasing inventory holding costs?

1.2 Main Contributions

The project makes four contributions. First, a leakage-audited, fully reproducible three-class delay-classification pipeline on a public humanitarian logistics dataset, with every preprocessing decision logged and every reported number traceable to a committed artifact. Second, a multi-angle model evaluation covering effectiveness, efficiency, and stability, including the finding that a systematic hyperparameter search could not improve on the original configuration, locating the

performance ceiling in the information content of pre-shipment features rather than in model tuning. Third, an attribute-level SHAP attribution showing that delay risk concentrates in destination country and vendor identity, with the two attributes dominating different risk classes. Fourth, a demonstration that embedding the classifier's probabilities into a standard safety-stock formula produces a policy that simultaneously reduces buffer inventory by 24.6% and stockout-exposure events by 36% relative to the fixed-buffer rule on the held-out test set, evidence that a model whose classification metrics fall short of aspirational targets can still dominate the incumbent operational policy.

2. Literature Integration and Comparison with Prior Work

This section compares the project's final results against the prior work identified in Milestone 2, explaining agreements and differences through the lens of data, features, and evaluation protocol rather than headline numbers alone.

2.1 Prior Work on the Same Dataset

Two prior studies use the same USAID SCMS dataset. Hanan (2025) addresses a regression task (predicting freight cost) and reports that XGBoost outperformed linear regression and random forest controls (RMSE $\approx$ USD 6,826) after one-hot encoding produced a very high-dimensional feature space. Although the target differs, the study independently corroborates two of this project's design outcomes on the identical data: gradient boosting handles the dataset's dominant categorical structure better than linear models, and vendor, country, and shipment-mode indicators carry the strongest signal. Robinson (n.d.) is closer in intent: a binary delayed/on-time classifier on the same records, reporting test F1 scores of 0.823-0.838 for XGBoost and CatBoost after SMOTE oversampling, with shipment mode, vendor, processing time, and country identified as the leading drivers.

The apparent gap between Robinson's binary F1 ($\approx$ 0.83) and this project's macro-F1 (0.60) is explained almost entirely by protocol rather than model quality. First, the binary formulation merges the intermediate at-risk band into a single late class, removing precisely the boundary this project found hardest (at-risk sits between on-time and delayed in feature space); a three-class macro average is penalized three times by its weakest class, while a binary score is not. Second, Robinson's feature set includes processing-time variables derived from the PQ and PO date fields, which this project excluded by design in Milestone 2 to keep every predictor available at the earliest planning moment. Third, this project's macro averaging weights the 4.8% delayed class equally with the 88.5% majority, whereas conventional F1 on a rebalanced binary problem is lifted by the majority class. When the comparison is restricted to a like-for-like signal, which attributes matter, the three studies converge on the same answer: vendor, country, and mode.

2.2 Delay-Prediction Literature Beyond the Dataset

Viellechner and Spinler (2020) built delay-prediction models for container shipping from 315 explanatory variables, obtaining 77% accuracy for their best models against a 59% naive baseline. Their improvement over the naive rule (+18 percentage points of accuracy) is proportionally comparable to this project's improvement over its baseline (+0.22 macro-F1, a 60% relative gain), and their conclusion that the business value of delay prediction lies in feeding schedule buffers and operational planning rather than in the classification metric itself anticipates this project's RQ3 result. Their richer feature space (including real-time vessel positions) also illustrates why this project's ceiling is informational: with only static, order-time attributes, some delay causes (weather, port congestion, customs events) are simply not observable at prediction time.

2.3 Methods Literature

Three results from the method literature were tested directly in this project. Chen and Guestrin's (2016) claim that regularized gradient boosting dominates on tabular data is confirmed: XGBoost beat the linear baseline on all 15 repeated-CV folds. Chawla et al.'s (2002) SMOTE improved the linear baseline (+0.053 macro-F1 in Milestone 3) exactly as the technique intends, but degraded the boosted-tree champion (−0.11), consistent with the caution that interpolated synthetic points are unrepresentative in high-dimensional, predominantly binary feature spaces; class weighting was therefore retained for the champion. Lundberg and Lee's (2017) SHAP framework provided the attribution layer for RQ1, and the standard safety-stock formulation under lead-time variability (Silver et al., 1998) provided the operational template for RQ3.

3. Methodology

This section summarizes the full pipeline, data preparation, feature engineering, model development, and the evaluation design, with emphasis on what changed between the Milestone 3 proof of concept and this final analysis. Complete implementation detail is in the repository: notebooks 02–05 (Milestone 3 stages) and notebooks 06–07 (Milestone 4 stages), each importing its logic from a documented module in `src/`.

3.1 Data Preparation and Leakage Control (Milestone 3, unchanged)

The three-class target was derived from the difference between the delivered-to-client and scheduled delivery dates: on-time (≤ 0 days), at-risk (1–14 days), delayed (> 14 days), yielding a 9,138 / 691 / 495 distribution (88.5% / 6.7% / 4.8%, imbalance ratio $\approx 18.5:1$). Weight and freight-cost fields stored as text with placeholder strings were repaired by deterministic cross-reference resolution, lifting usable coverage from 61.7% / 60.0% to 84.2% / 82.7%. Twenty-one columns were removed with a written rationale each (three leakage fields unavailable at order placement, four identifiers, fourteen excluded by the Milestone 2 feature design). The data was split 80/20 with stratification (8,259 train / 2,065 test rows; `random_state = 42`; exact row identifiers committed in `split_indices.json`), and all statistics were fitted on the training partition only.

3.2 Feature Engineering (Milestone 3, unchanged)

A FeatureEngineer class with a strict fit/transform contract produced 154 features: mode-stratified median imputation with informative-missingness flags, log1p transformation of the three right-skewed monetary and weight fields, chi-squared pre-selection of six categorical attributes on the training split (all retained, $\alpha = 0.05$), one-hot encoding (142 columns), a vendor risk tier thresholded at the network-wide on-time rate, and five mode $\times$ tier interactions. The Milestone 4 analysis deliberately holds this feature pipeline fixed so that any change in results is attributable to the modelling and decision layers.

3.3 Final Model Development (Milestone 4)

Three extensions were made, implemented in src/final_models.py and Notebook 06.

Hyperparameter search: 20 configurations were sampled with a fixed seed from a declared grid spanning 300-900 trees, learning rates 0.05-0.10, depths 4-8, minimum child weights 1-5, subsampling and column-sampling rates 0.8-1.0, and L2 regularization 1-5, bracketing the Milestone 3 defaults on both sides so the search could confirm or reject them. Each configuration was scored by the same stratified 5-fold protocol and class-weight strategy as Milestone 3, with early stopping (patience 50) monitored on a 10% stratified carve-out of each fold's training portion. The loop is written explicitly rather than through library search utilities, keeping per-fold weighting and stopping auditable under the project's no-black-box rule.

Repeated cross-validation: the best sampled configuration, the Milestone 3 default configuration, and the logistic regression baseline were then compared on identical folds across three independent 5-fold repeats (15 paired scores per model).

Champion selection criterion (stated before any test-set contact): the XGBoost arm with the higher repeated-CV mean macro-F1 becomes the final champion; if the arms sit within one fold-level standard deviation, the cheaper configuration is preferred on parsimony grounds.

Deterministic final fit: the selected champion and the baseline were refit on the full training split with a single thread, a fixed seed, and a fixed tree count (no early stopping). This removes two Milestone 3 artifacts at once: the run-to-run metric variation caused by thread-order effects in multithreaded histogram training, and the 10% of training data previously sacrificed to the early-stopping carve-out in the final fit. Notebook 06 verifies determinism directly; two independent refits produce bit-identical predictions.

3.4 Evaluation Design

The evaluation covers three angles.

Effectiveness: macro-averaged F1 (primary) and macro-averaged AUPRC on the held-out test set, with per-class precision/recall/F1 and the confusion matrix; accuracy is reported only for context because a constant on-time prediction would score 88.5%.

Efficiency: wall-clock cost of the one-off hyperparameter search, final training time, per-shipment scoring latency, and model artifact size, all measured single-threaded.

Stability: fold-level dispersion across the 15 repeated-CV folds, the CV-to-test generalization gap, and the determinism check. The champion–baseline claim is supported by a one-sided Wilcoxon signed-rank test on the 15 paired per-fold differences (direction stated in advance in RQ2; the non-parametric test is retained because fold-level F1 differences cannot be assumed normal).

Two decision-layer analyses follow the same leakage discipline as the rest of the pipeline. The cost-sensitive decision rule (class-probability weight multipliers replacing plain argmax) is selected to minimize expected cost on out-of-fold training predictions only, then applied once to the test set; an illustrative asymmetric cost matrix prices a missed at-risk warning at five times a false alarm and a missed severe delay at ten times, with sensitivity to this assumption reported. The safety-stock simulation implements the Milestone 1 formula with a class-mixture standard deviation: class-conditional lateness statistics (lateness = $\max(\text{delay days}, 0)$, since early arrival is not a stockout risk) are learned from the training split and combined with each test shipment's predicted probabilities; $z = 1.645$ (95% cycle-service level) for both policies; and because the dataset contains no demand quantities, buffers are expressed in days of cover, which leaves the fixed-versus-dynamic comparison unaffected.

4. Results: Multi-Angle Model Evaluation

4.1 Hyperparameter Search: The Defaults Were Validated

The search produced a clear negative-space finding: across 20 configurations, no candidate improved on the Milestone 3 defaults beyond fold-level noise (Table 1). The best sampled configuration (900-tree budget, learning rate 0.05, depth 6; early stopping used a mean of 885 rounds) reached a 5-fold CV macro-F1 of 0.5799 ± 0.0170 , statistically indistinguishable from the default configuration's 0.5803 ± 0.0169 , and lower-capacity or heavily regularized candidates fell further behind. The Milestone 3 concern that the exhausted 500-round budget signalled untapped headroom is therefore resolved in the opposite direction: additional capacity buys nothing, because the ceiling is set by what pre-shipment attributes can reveal, not by the model configuration. This finding redirected improvement effort toward the decision layer (Section 5) rather than further model complexity.

Config	Trees (used)	LR	Depth	Subsample / Colsample	Reg λ	CV macro-F1 ($\pm$ SD)
Best sampled (#15)	900 (885)	0.05	6	1.0 / 1.0	1.0	0.5799 ± 0.0170
2nd (#14)	900 (848)	0.05	6	0.8 / 0.8	1.0	0.5789 ± 0.0169

Config	Trees (used)	LR	Depth	Subsample / Colsample	Reg λ	CV macro-F1 ($\pm$ SD)
3rd (#19)	900 (320)	0.10	8	0.8 / 0.8	5.0	0.5719 $\pm$ 0.0178
M3 default (reference)	500	0.10	6	1.0 / 1.0	1.0	0.5803 $\pm$ 0.0169

Table 1. Hyperparameter search: top three of 20 sampled configurations against the Milestone 3 default (stratified 5-fold CV, class-weight strategy, training split only). Full grid in `outputs/tables/hyperparameter_tuning_results.csv`.

4.2 Stability: Repeated Cross-Validation and Determinism

Across 3×5 repeated stratified CV on identical folds (Table 2, Figure 1), the Milestone 3 default configuration achieved the highest mean macro-F1 (0.5939 ± 0.0252), narrowly ahead of the tuned candidate (0.5877 ± 0.0240) and far ahead of the baseline (0.3713 ± 0.0067). The two XGBoost arms overlap almost completely, visual confirmation that tuning moved nothing, while the ranking against the baseline is preserved in every one of the 15 folds. The baseline's very small dispersion is characteristic of a high-bias model: it is stable, but stably poor. Model behaviour is also exactly reproducible: the determinism check in Notebook 06 confirms that two independent refits of the final champion produce bit-identical predictions, eliminating the small run-to-run variation observed in Milestone 3.

Model	Macro-F1 (mean $\pm$ SD)	AUPRC (mean $\pm$ SD)	Mean fit time (s)
XGBoost (M3 default)	0.5939 ± 0.0252	0.6004 ± 0.0282	2.94
XGBoost (tuned candidate)	0.5877 ± 0.0240	0.5948 ± 0.0290	5.18
Logistic Regression (baseline)	0.3713 ± 0.0067	0.4256 ± 0.0076	0.30

Table 2. Repeated stratified cross-validation summary (3 repeats $\times$ 5 folds, training split only).

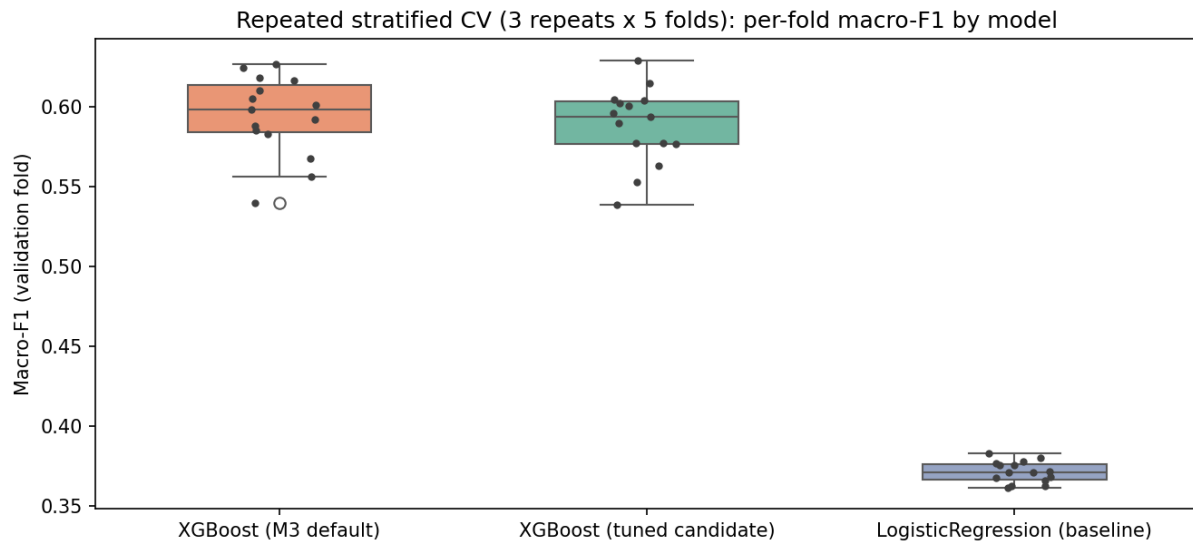


Figure 1. Per-fold macro-F1 across 15 repeated-CV folds. The two XGBoost arms overlap; the baseline is stable but far below.

4.3 Champion Selection

By the pre-stated criterion, the Milestone 3 default configuration was retained as the final champion: it has the higher repeated-CV mean, and, since the two arms sit within one standard deviation of each other, it would also have been preferred as the cheaper configuration (2.94 s versus 5.18 s mean fit time). The selection is therefore not one metric alone but a convergence of performance, parsimony, and cost, and it was made entirely on training-split evidence before any test-set contact. The value the search added is confidence: the deployed configuration is now known to be at or near the ceiling of its family, rather than assumed to be.

4.4 Statistical Comparison on Fifteen Paired Folds

The one-sided Wilcoxon signed-rank test on the 15 paired per-fold macro-F1 differences rejects the null hypothesis decisively: $W = 120$, $p = 3.05 \times 10^{-5}$ (Table 3). Every one of the 15 folds favours the champion, with a mean advantage of +0.223 macro-F1 and a worst-fold advantage of +0.168. This resolves the Milestone 3 limitation in full: the earlier test rested on five pairs with its p-value at the one-sided attainable floor (0.03125), whereas the present result sits four orders of magnitude below the $\alpha = 0.05$ threshold with a minimum attainable p of ≈ 0.00003 .

Test	n pairs	W	p-value (one-sided)	Mean diff.	Min. diff.
Wilcoxon signed-rank (champion – baseline)	15	120	3.05×10^{-5}	+0.223	+0.168

Table 3. Wilcoxon signed-rank test on paired per-fold macro-F1 differences across the 15 repeated-CV folds.

4.5 Effectiveness on the Held-Out Test Set

The deterministic final models were evaluated once against the 2,065-row test set (Table 4). The champion reached a macro-F1 of 0.5992 and macro-AUPRC of 0.5942, against 0.3779 and 0.4388 for the baseline. The champion's test score sits just +0.005 above its repeated-CV mean, well inside one-fold level standard deviation (0.0252), so cross-validation is an honest estimator of generalization.

The improvement over the Milestone 3 test result (0.5863) deserves explanation, because the model configuration is identical. The cause is the final-fit protocol: the Milestone 3 champion reserved a 10% early-stopping carve-out even in its final fit, training on only 90% of the training split, whereas the deterministic Milestone 4 fit uses a fixed tree count and therefore all 8,259 training rows. Returning roughly 826 rows, disproportionately valuable for classes with only hundreds of examples, lifted at-risk F1 from 0.316 to 0.362 while the other classes held level (on-time 0.927 to 0.926, delayed 0.515 to 0.510; Table 5, Figure 2): the scarcest decision boundary absorbed nearly all of the benefit of the returned data. The gain is a data effect, not a tuning effect, and it is consistent with the informational-ceiling interpretation of Section 4.1.

Model	Test macro-F1	Test AUPRC (macro)	AUPRC (Delayed)	Repeated-CV mean	Gap
XGBoost (final champion)	0.5992	0.5942	0.4929	0.5939	+0.0053
Logistic Regression (baseline)	0.3779	0.4388	0.2071	0.3713	+0.0066

Table 4. Held-out test performance versus repeated-CV estimates.

Class	Precision	Recall	F1	F1 (Milestone 3)	Support
On-time (0)	0.939	0.914	0.926	0.927	1,828
At-risk (1)	0.322	0.413	0.362	0.316	138
Delayed (2)	0.486	0.535	0.510	0.515	99
Macro average	0.582	0.621	0.599	0.586	2,065

Table 5. Final champion per-class performance on the held-out test set (default argmax decision rule), with Milestone 3 F1 for comparison.

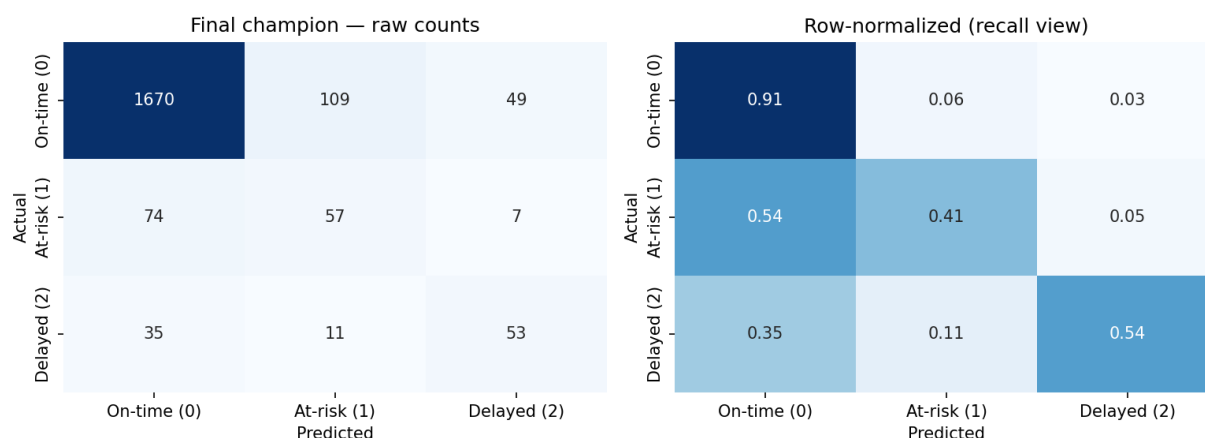


Figure 2. Final champion confusion matrices on the held-out test set: raw counts (left) and row-normalized recall view (right).

The error structure retains its Milestone 3 shape. The majority class is predicted reliably; severe delays carry recognisable vendor, route, and freight signatures and are recovered at 53.5% recall; and the at-risk band remains the hardest, because shipments that will run 1–14 days late are often indistinguishable at order placement from on-time ones. That intermediate band is addressed operationally, rather than by more modelling, in Section 5.2.

4.6 Efficiency

All costs were measured on a single CPU thread, the deterministic configuration, and therefore the worst case (Table 6). The recurring costs are trivial for a dataset of this size: the final champion trains in 3.61 seconds, scores the entire test set in 0.07 seconds (0.032 ms per shipment), and occupies 3.60 MB on disk. The one-off hyperparameter search cost 6.9 minutes of wall-clock time; given that it returned no better configuration, its value was diagnostic rather than performance-improving, and the negative result means future retraining cycles can skip the search entirely. The baseline is an order of magnitude cheaper to train, but Section 4.4 shows the champion's +0.223 macro-F1 is bought for under four seconds — an unambiguous trade at this scale. Nothing in the pipeline requires more than commodity hardware, which matters for the NGO deployment context of the beneficiary.

Measure	Value
One-off hyperparameter search (20 configs × 5 folds, single thread)	6.9 min
Final champion fit (full training split, single thread)	3.61 s
Final baseline fit	0.30 s
Champion scoring, full 2,065-row test set	0.066 s

Measure	Value
Champion scoring per shipment	0.032 ms
Champion artifact size	3.60 MB
Baseline artifact size	0.01 MB

Table 6. Efficiency measurements (Notebook 07, single-threaded).

5. Interpretation: Answers to the Research Questions

5.1 RQ1 – Which Pre-Shipment Attributes Drive Delay Risk?

Answer: Destination country and vendor identity are the two dominant predictors, followed by the shipment's physical and financial profile (freight cost, weight, quantity, line-item value); shipment mode and commodity form contribute a secondary signal.

Evidence: TreeExplainer SHAP values were computed on the full test set and aggregated from the 154 model columns back to the original shipment attributes (Figure 3, full table in outputs/tables/shap_grouped_importance.csv). Country carries the largest overall attribution (0.635), vendor is second (0.594), and the two dominate different classes: vendor is the leading driver of the on-time and at-risk distinctions (0.890 and 0.631), while country is by far the leading driver of the delayed class (0.903 versus 0.263 for vendor), severe delays are a route phenomenon (customs, port, and last-mile conditions at the destination), while marginal lateness is a supplier phenomenon.

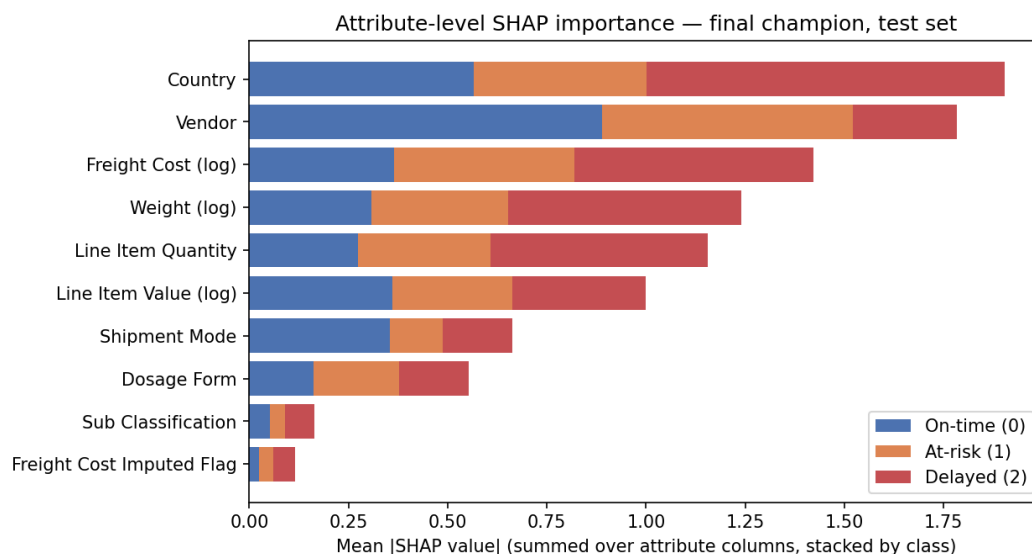


Figure 3. Attribute-level SHAP importance (mean |SHAP| summed over each attribute's columns, stacked by class), final champion on the held-out test set.

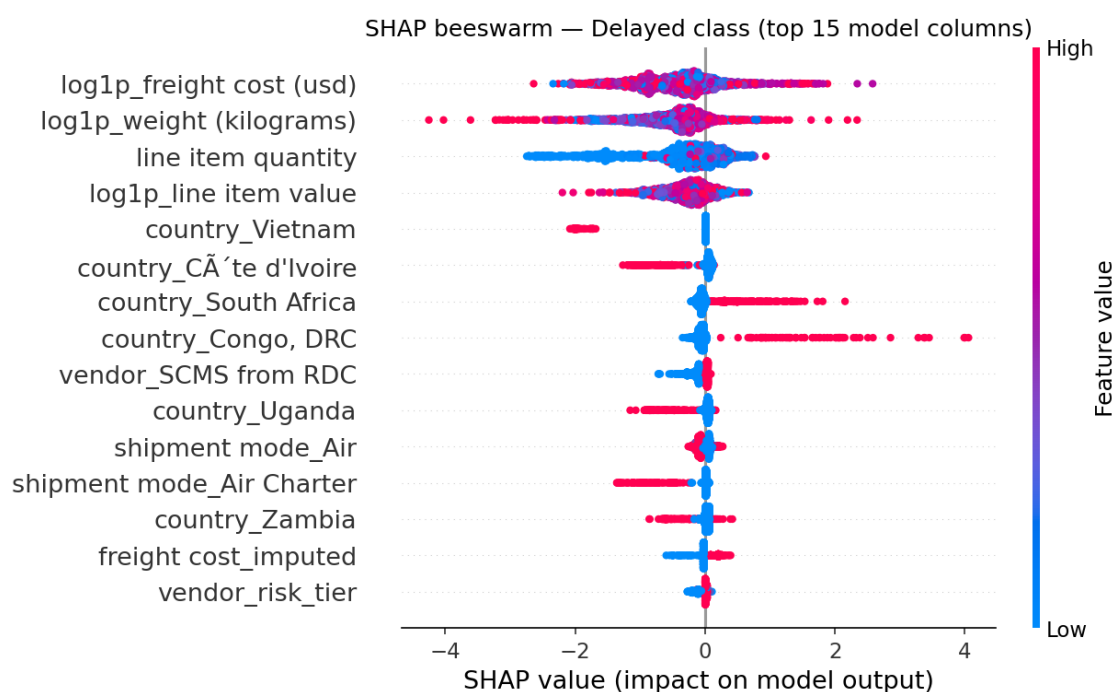


Figure 4. SHAP beeswarm for the Delayed class: top 15 individual model columns.

The column-level view (Figure 4) makes the drivers concrete: specific destination countries, the log-freight and log-weight profile of heavy consolidated shipments, and individual vendor indicators push predictions toward the delayed class. Two secondary observations complete the picture. First, the fulfil-via channel receives zero SHAP attribution despite passing the chi-squared screen, its signal is fully absorbed by the correlated vendor and country columns, a reminder that univariate association and marginal contribution in a multivariate model are different questions. Second, the Milestone 3 vendor-tier finding is confirmed and sharpened: seven of 69 training vendors, handling 68.5% of volume, carry a 16.2% combined delay rate against 1.3% for the rest, and the SHAP ranking shows the model leans on exactly this vendor concentration.

Practical meaning: the Milestone 1 interpretability criterion, at least three actionable drivers requiring no new data collection, is met. A procurement director can: (1) prioritise SLA renegotiation and dual-sourcing for the small set of high-volume, high-risk vendors; (2) apply route-level planning buffers and customs pre-clearance effort for the high-attribution destination countries; and (3) treat heavy, high-freight consolidated shipments as a distinct risk category when choosing mode and dispatch timing.

5.2 RQ2 – Does the Champion Meaningfully Beat the Baseline, and Can Its Decisions Be Made Cost-Aware?

Answer: Yes, decisively and stably. The champion exceeds the baseline by +0.221 macro-F1 on the held-out test set (0.5992 versus 0.3779) and by +0.223 on average across 15 repeated-CV folds, winning every fold; the one-sided Wilcoxon test returns $p = 3.05 \times 10^{-5}$. The difference is

not marginal in operational terms either: the baseline's delayed-class AUPRC (0.207) is barely above the class prevalence, meaning the linear model cannot rank severe-delay risk usefully, while the champion's 0.493 supports genuine triage.

The champion's remaining weakness, at-risk recall of 0.413 under the default argmax rule, was addressed at the decision layer. Under the illustrative 5:1 cost asymmetry, the expected-cost-minimizing weights selected on out-of-fold training predictions were (1.0, 2.0, 4.0). Applied once to the test set (Table 7), the rule raises at-risk recall from 0.413 to 0.493 and delayed recall from 0.535 to 0.687, over two-thirds of severe delays are now flagged, at the price of on-time recall (0.914 → 0.834) and at-risk precision (0.322 → 0.268), reducing expected cost per shipment on the held-out test set by 7.4% (0.4392 → 0.4068). Macro-F1 falls from 0.599 to 0.559 under this rule, which is the point of the analysis: once error costs are asymmetric, the F1-optimal operating point is no longer the cost-optimal one, and the two objectives must be kept distinct.

Class	Recall (default)	Recall (cost-sens.)	Precision (default)	Precision (cost-sens.)
On-time (0)	0.914	0.834	0.939	0.952
At-risk (1)	0.413	0.493	0.322	0.268
Delayed (2)	0.535	0.687	0.486	0.324
Macro average	0.621	0.671	0.582	0.514

Table 7. Default argmax versus cost-sensitive decision rule on the held-out test set (weights 1.0 / 2.0 / 4.0, selected on out-of-fold training predictions under the 5:1 cost matrix).

Because the 5:1 ratio is an assumption, its sensitivity was quantified: under a 2:1 ratio the cost-minimizing search selects the default argmax rule itself (weights 1.0 / 1.0), while under 10:1 it pushes at-risk recall to 0.623 at precision 0.213. The deliverable to the beneficiary is therefore not one threshold but a dial: the organization chooses the operating point that matches its true cost of a missed late shipment relative to a wasted review, and the machinery for translating that ratio into a decision rule is committed in the repository.

5.3 RQ3 – Does the Dynamic Safety-Stock Formula Beat the Fixed Buffer?

Answer: Yes, on the held-out test set the dynamic policy dominates the fixed policy on both objectives simultaneously.

Evidence: the fixed policy applies $z \times \sigma$ of network-wide training lateness ($1.645 \times 12.07 \approx 19.85$ days of cover) to every shipment, accumulating 41,000 buffer-days across the 2,065 test shipments while still leaving 78 stockout-exposure events (3.78%), shipments whose actual lateness exceeded the uniform buffer. The dynamic policy computes each shipment's buffer from

its predicted class-mixture lateness standard deviation and holds 30,895 buffer-days, 24.6% less inventory, while stockout-exposure events fall to 50 (2.42%), a 36% reduction (Table 8, Figure 5).

Policy	Total buffer-days held	Stockout-exposure events	Exposure rate
Fixed (19.85 days for all)	41,000	78	3.78%
Dynamic (model-driven)	30,895 (-24.6%)	50 (-36%)	2.42%

Table 8. Safety-stock policy comparison on the held-out test set ($z = 1.645$; buffers in days of cover; lateness = $\max(\text{delay days}, 0)$).

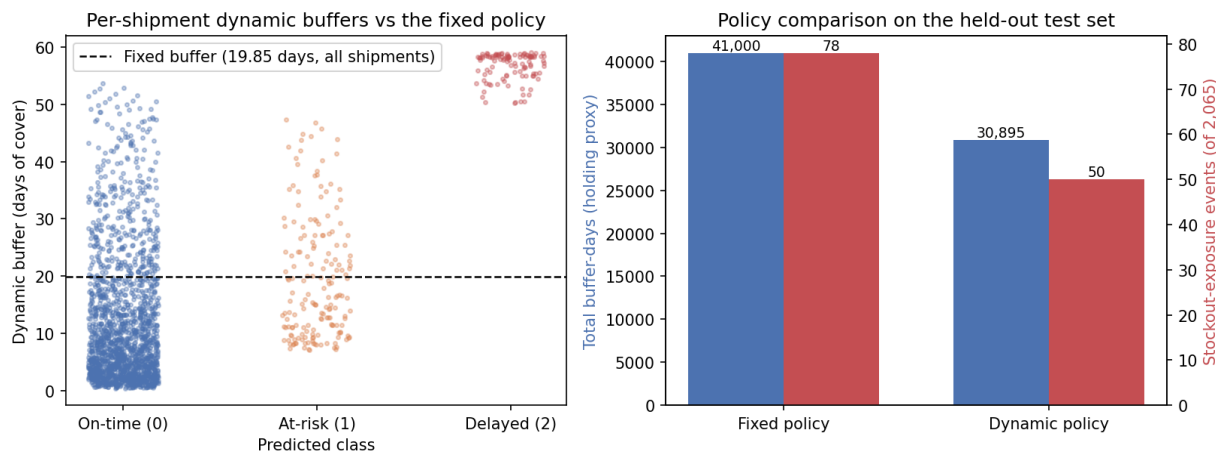


Figure 5. Left: per-shipment dynamic buffers by predicted class against the uniform fixed buffer. Right: totals for both policies on the held-out test set.

The Milestone 1 directional success criterion is met with room to spare: shipments predicted on-time receive a mean buffer of 11.9 days (median 7.9) versus the fixed 19.85, the source of the holding-cost saving, while shipments predicted delayed receive a mean of 56.7 days, nearly triple the fixed buffer, which is where the stockout protection comes from. The mechanism explains why a classifier with macro-F1 of 0.60 can still dominate the incumbent policy: the fixed rule wastes buffer on the 88.5% of shipments that arrive on time and under-protects the 4.8% that arrive very late, so even imperfect risk ranking reallocates inventory in the right direction. Because the dataset contains no demand quantities, both policies are expressed per unit of average daily demand (days of cover); scaling by real demand would change the absolute totals but not the comparison.

5.4 Technical Performance versus Stakeholder Value

Table 9 closes the loop on the measurable success criteria committed in Milestone 1, and the honest scorecard is mixed in an instructive way. The two aspirational classification targets were not met: test macro-F1 reached 0.5992 against the 0.72 target, and delayed-class AUPRC reached 0.4929 against 0.70. Those targets were set before data profiling revealed the at-risk

band's intrinsic ambiguity, and the evidence of Sections 4.1-4.2, a systematic search that could not beat the defaults, and a data-limited final-fit gain, indicates the shortfall is informational, not methodological: order-time attributes do not observe weather, customs events, or port congestion. Both operational criteria, by contrast, were met in full. This is the report's central distinction between technical performance and business value: the classification metric measures distance from a perfect oracle, while the stakeholder decision only requires beating the incumbent fixed-buffer policy and unranked triage, a much lower bar that the system clears decisively.

Milestone 1 success criterion	Target	Achieved	Status
Test macro-F1 (three-class)	> 0.72	0.5992	Not met
Test AUPRC, Delayed class	> 0.70	0.4929	Not met
Statistically meaningful improvement over baseline	Required	+0.223, $p = 3.05 \times 10^{-5}$	Met
Directionally correct buffer reallocation (lower for on-time, higher for delayed)	Required	11.9 vs 19.85 vs 56.7 days; -24.6% holding, -36% exposure	Met (dominates)
≥ 3 actionable drivers identified via SHAP	Required	Vendor, country/route, freight-weight profile (+ mode)	Met

Table 9. Scorecard against the measurable success criteria defined in Milestone 1.

6. Ethics, Privacy, Bias, and Limitations

6.1 Privacy

The dataset contains no personally identifiable information: records describe organizational shipments (vendors, countries, commodities, costs), and the privacy audit in Milestone 1 confirmed the data was de-identified before public release by USAID under a CC-BY licence. The residual privacy consideration is organizational rather than personal; vendor names are real companies, and the vendor risk tier effectively publishes a reputational judgement. This report therefore reports tier-level aggregates only and does not name individual high-risk vendors; an operational deployment should treat vendor risk scores as confidential procurement inputs, not public ratings.

6.2 Bias and Fairness Risks

Two bias mechanisms are realistic for this system. First, destination country is the model's strongest attribution for severe delays, so the dynamic policy systematically assigns larger buffers and, implicitly, higher scrutiny, to shipments bound for particular low and middle-income countries. The delay signal is real in the historical data, but its causes (customs capacity, port infrastructure)

are conditions of the destination environment, not qualities of the recipient programs; if risk scores were ever used to deprioritize rather than protect those destinations, the model would compound existing disadvantage. The appropriate use, consistent with the beneficiary framing, is protective: higher predicted risk should trigger larger buffers and earlier dispatch, never reduced service. Second, a feedback-loop risk: if planners act on predictions (for example, switching away from flagged vendors), future training data will reflect those interventions, and naive retraining would learn a distorted picture. Any retraining cycle should log interventions so treated shipments can be identified.

6.3 Limitations

- **Temporal validity.** The data spans 2006-2015; the SCMS vendor roster and freight network no longer exist in that form. The structural relationships (route and vendor concentration of risk) are plausibly durable, but the specific coefficients are not, and no deployment claim is made without retraining on current data.
- **At-risk ambiguity.** Even after the decision-layer improvement, roughly half of at-risk shipments are missed, and three of four at-risk alarms are false. The class is intrinsically hard to separate at order time, and conclusions about the 1-14-day band should carry that uncertainty.
- **Probability calibration.** Class-weighted training distorts predicted probabilities, and both the cost-sensitive rule and the safety-stock mixture consume those probabilities. Rankings are unaffected, but the absolute buffer sizes and expected costs would shift under calibration, the first item of future work.
- **Cost-matrix assumption.** The 5:1 asymmetry is illustrative. The sensitivity analysis bounds its effect, but a deployed rule requires the organization's actual cost ratio.
- **Demand data absence.** Buffers are expressed in days of cover; absolute inventory quantities require demand-rate data the dataset does not contain.
- **Statistical scope.** The 15 repeated-CV pairs share underlying data across repeats, so they are not fully independent; the Wilcoxon result should be read as strong evidence within this dataset, not as a claim about all supply-chain data.
- **Unobserved disruption factors.** The dataset contains only order-time operational attributes and does not capture weather disruptions, customs delays, port congestion, labour disputes, transportation infrastructure failures, or geopolitical events. Consequently, some delay events may remain inherently unpredictable regardless of algorithm choice, placing an upper bound on achievable predictive performance.

These limitations shape the recommended actions directly: the system is fit for decision support, ranking shipments for review and sizing relative buffers, and is not fit for unsupervised automation of procurement decisions, vendor deselection, or public vendor scoring.

7. Recommendations for the Next Phase

Each recommendation extends a measured weakness rather than adding another algorithm for its own sake.

- **Probability calibration:** Apply isotonic or Platt calibration on out-of-fold predictions before the safety-stock mixture, then re-run the Table 8 simulation; this targets the calibration limitation and would make buffer sizes trustworthy in absolute terms.
- **Ordinal reformulation:** The three classes are ordered, and the confusion structure (at-risk confused with its neighbours, almost never on-time with delayed) suggests an ordinal or two-stage model could recover part of the at-risk boundary that nominal classification wastes.
- **Cost elicitation:** Replace the illustrative 5:1 matrix with ratios elicited from procurement stakeholders (emergency airfreight premium versus review cost), converting the Section 5.2 dial into a defensible operating point.
- **Temporal external validation:** Train on 2006-2012 and test on 2013-2015 to measure how quickly route and vendor signal decays, directly addressing the temporal-validity limitation and informing a retraining cadence.
- **Demand integration:** Join facility-level consumption data where available so buffers convert from days of cover to units, completing the inventory-cost accounting that the current dataset cannot support.

8. Conclusion

This capstone set out to test whether delay risk in a global health supply chain can be predicted from order-time attributes and turned into a better inventory policy. The final evidence answers all three research questions. Risk is concentrated and identifiable: destination country and vendor identity dominate the SHAP attribution, with severe delays behaving as a route phenomenon and marginal lateness as a supplier phenomenon (RQ1). The boosted-tree champion beats the linear baseline by +0.223 macro-F1 across every one of 15 repeated cross-validation folds ($p = 3.05 \times 10^{-5}$), generalizes to unseen data within one fold-level standard deviation, trains in seconds, and is bit-for-bit reproducible (RQ2). And when its probabilities are embedded in the standard safety-stock formula, the resulting dynamic policy holds 24.6% less buffer inventory while cutting stockout exposure by 36% relative to the fixed rule (RQ3). The classification metrics fell short of the aspirational Milestone 1 targets, and the hyperparameter search demonstrated that the shortfall is informational rather than methodological, but the project's most useful lesson runs through exactly that gap: for the procurement planner the system was built to serve, the relevant question was never whether the model is perfect, but whether it beats the policy currently in use. It does, on both cost and risk, at negligible computational expense, with every number in this report traceable to a committed artifact in the public repository:

<https://github.com/MA-Gondal/CIND-820-Capstone-Project>

9. References

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10. AI Use Declaration (TMU Policy 60)

In accordance with TMU Academic Integrity Policy 60, generative AI (Claude, Anthropic) was used in this milestone for: drafting the Milestone 4 code modules (`src/final_models.py`, `src/impact.py`) and Notebooks 06–07 that implement the analysis agenda set out in the Milestone 3 report; discussing methodological decisions (the champion selection criterion, leakage-free selection of the cost-sensitive decision weights, the class-mixture formulation of the safety-stock variance); and drafting documentation and report prose. Verification followed the project's established protocol: every notebook was executed locally end-to-end, all reported numbers were cross-checked between notebook output and the saved CSV artifacts committed in `outputs/tables/`, the Notebook 06 determinism check was confirmed, prose was reviewed and edited into the author's own words, and every cited source was personally accessed. AI was not used to fabricate results, sources, or interpretations, and the author can explain and defend every design decision independently. The complete per-activity usage and verification log is committed in `ai_use_declaration.md` in the repository.